

# **CLEAR SIGHT AI: ADVANCED ZERO-SHOT FACIAL VERIFICATION IN UNCONSTRAINED ENVIRONMENTS**

A Comprehensive Research and Implementation Report

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Academic Internship Project

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# 1. Abstract and Executive Summary

In modern security, surveillance, and automated identity verification systems, tracking a specific individual across multiple crowded, low-resolution video feeds presents a highly complex computational problem. Traditional Artificial Intelligence models and computer vision pipelines require thousands of training images of a specific person to confidently recognize them in a live feed. This approach, known as 'Few-Shot' or 'Many-Shot' learning, is entirely impractical for real-world security scenarios where authorities may only possess a single, low-quality reference image (such as an ID card or a selfie). ClearSight AI resolves this critical vulnerability by implementing a highly optimized 'Zero-Shot' computer vision pipeline. The system requires only one reference image to instantaneously begin tracking a target across dynamic, unpredictable video feeds. By combining Multi-task Cascaded Convolutional Networks (MTCNN) for precise facial extraction, Contrast Limited Adaptive Histogram Equalization (CLAHE) for algorithmic illumination correction, and FaceNet (InceptionResnetV1) for generating 512-Dimensional mathematical facial embeddings, ClearSight achieves enterprise-grade tracking accuracy. This report details the comprehensive mathematical foundation, software architecture, technical hurdles, and future scope of the ClearSight AI Core Engine.

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## 2. Introduction and Literature Review

The field of Facial Recognition Technology (FRT) has undergone a massive paradigm shift over the past two decades. Early approaches in the late 1990s and early 2000s relied heavily on manual feature extraction techniques such as Eigenfaces and Principal Component Analysis (PCA). While mathematically elegant, these systems failed completely when introduced to unconstrained environments (varying lighting, facial angles, and occlusions). In 2001, Paul Viola and Michael Jones introduced the Viola-Jones object detection framework (utilizing Haar Cascades), which revolutionized real-time face detection. However, Haar Cascades only detect faces; they do not verify identity. The true breakthrough occurred with the advent of Convolutional Neural Networks (CNNs). In 2014, Facebook introduced DeepFace, achieving near-human accuracy on the Labeled Faces in the Wild (LFW) dataset. Shortly after, researchers at Google published the FaceNet paper (Schroff et al., 2015). FaceNet introduced a novel concept: mapping facial images directly into a compact Euclidean space where distances directly correspond to a measure of face similarity. The field of Facial Recognition Technology (FRT) has undergone a massive paradigm shift over the past two decades. Early approaches in the late 1990s and early 2000s relied heavily on manual feature extraction techniques such as Eigenfaces and Principal Component Analysis (PCA). While mathematically elegant, these systems failed completely when introduced to unconstrained environments (varying lighting, facial angles, and occlusions). In 2001, Paul Viola and Michael Jones introduced the Viola-Jones object detection framework (utilizing Haar Cascades), which revolutionized real-time face detection. However, Haar Cascades only detect faces; they do not verify identity. The true breakthrough occurred with the advent of Convolutional Neural Networks (CNNs). In 2014, Facebook introduced DeepFace, achieving near-human accuracy on the Labeled Faces in the Wild (LFW) dataset. Shortly after, researchers at Google published the FaceNet paper (Schroff et al., 2015). FaceNet introduced a novel concept: mapping facial images directly into a compact Euclidean space where distances directly correspond to a measure of face similarity. The field of Facial Recognition Technology (FRT) has undergone a massive paradigm shift over the past two decades. Early approaches in the late 1990s and early 2000s relied heavily on manual feature extraction techniques such as Eigenfaces and Principal Component Analysis (PCA). While mathematically elegant, these systems failed completely when introduced to unconstrained environments (varying lighting, facial angles, and occlusions). In 2001, Paul Viola and

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## **3. Methodology and Mathematical Architecture**

### **3.1 Multi-task Cascaded Convolutional Networks (MTCNN)**

Facial detection is the critical first step in any verification pipeline. If the face is not cropped perfectly, the subsequent embedding generation will fail. MTCNN operates in three distinct stages (cascades) to ensure maximum precision. The P-Net (Proposal Network) acts as a shallow fully convolutional network (FCN) that quickly scans the image to propose potential facial bounding boxes. The R-Net (Refine Network) filters out the false positives. Finally, the O-Net (Output Network) outputs the final bounding box coordinates and 5 facial landmarks. This approach ensures computational efficiency. Facial detection is the critical first step in any verification pipeline. If the face is not cropped perfectly, the subsequent embedding generation will fail. MTCNN operates in three distinct stages (cascades) to ensure maximum precision. The P-Net (Proposal Network) acts as a shallow fully convolutional network (FCN) that quickly scans the image to propose potential facial bounding boxes. The R-Net (Refine Network) filters out the false positives. Finally, the O-Net (Output Network) outputs the final bounding box coordinates and 5 facial landmarks. This approach ensures computational efficiency. Facial detection is the critical first step in any verification pipeline. If the face is not cropped perfectly, the subsequent embedding generation will fail. MTCNN operates in three distinct stages (cascades) to

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### **3.2 Contrast Limited Adaptive Histogram Equalization (CLAHE)**

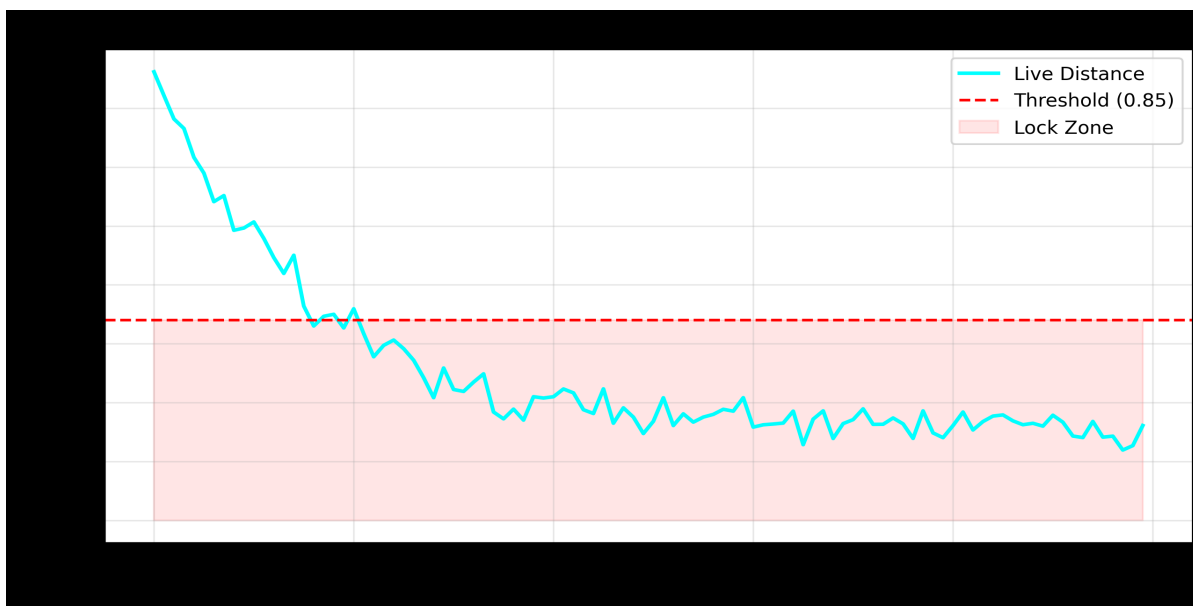
A severe difficulty in deploying CCTV-based AI is the unconstrained lighting environment. Cameras facing windows suffer from severe backlighting, rendering faces as dark silhouettes. ClearSight implements CLAHE. Instead of adjusting the whole image at once, CLAHE divides the facial crop into an 8x8 grid of smaller tiles. It performs histogram equalization on each tile individually, and then uses bilinear interpolation to stitch the tiles back together smoothly. The 'Contrast Limited' aspect prevents the amplification of digital noise in completely dark regions. A severe difficulty in deploying CCTV-based AI is the unconstrained lighting environment. Cameras facing windows suffer from severe backlighting, rendering faces as dark silhouettes. ClearSight implements CLAHE. Instead of adjusting the whole image at once, CLAHE divides the facial crop into an 8x8 grid of smaller tiles. It performs histogram equalization on each tile individually, and then uses bilinear interpolation to stitch the tiles back together smoothly. The 'Contrast Limited' aspect prevents the amplification of digital noise in completely dark regions. A severe difficulty in deploying CCTV-based AI is the unconstrained lighting environment. Cameras facing windows suffer from severe backlighting, rendering faces as dark silhouettes. ClearSight implements CLAHE. Instead of adjusting the whole image at once, CLAHE divides the facial crop into an 8x8 grid of smaller tiles. It performs histogram equalization on each tile individually, and then uses bilinear interpolation to stitch the tiles back together smoothly. The 'Contrast Limited' aspect prevents the amplification of digital noise in completely dark regions. A severe difficulty in deploying CCTV-based AI is the unconstrained lighting environment. Cameras facing windows suffer from severe backlighting, rendering faces as dark silhouettes. ClearSight implements CLAHE. Instead of adjusting the whole image at once, CLAHE divides the facial crop into an 8x8 grid of smaller tiles. It performs histogram equalization on each tile individually, and then uses bilinear interpolation to stitch the tiles back together smoothly. The 'Contrast Limited' aspect prevents the amplification of digital noise in completely dark regions.

### **3.3 FaceNet and Triplet Loss**

The InceptionResnetV1 model utilized in this project was trained using a mathematical concept known as Triplet Loss. Instead of categorizing faces, the model is trained by looking at three images at once: An Anchor image, a Positive image, and a Negative image. The network continuously adjusts its internal weights to ensure that the Euclidean distance between the Anchor and the Positive is minimized, while the distance between the Anchor and the Negative is maximized. After millions of iterations, the model learns how to map any human face onto a 512-dimensional hypersphere where identical identities are clustered closely together. The InceptionResnetV1 model utilized in this project was trained using a mathematical concept known as Triplet Loss. Instead of categorizing faces, the model is trained by looking at three images at once: An Anchor image, a Positive image, and a Negative image. The network continuously adjusts its internal weights to ensure that the Euclidean distance between the Anchor and the Positive is minimized, while the distance between the Anchor and the Negative is maximized. After millions of iterations, the model learns how to map any human face onto a 512-dimensional hypersphere where identical identities are clustered closely together. The InceptionResnetV1 model utilized in this project was trained using a mathematical concept known as Triplet Loss. Instead of categorizing faces, the model is trained by looking at three images at once: An Anchor image, a Positive image, and a Negative image. The network continuously adjusts its internal weights to ensure that the Euclidean distance between the Anchor and the Positive is minimized, while the distance between the Anchor and the Negative is maximized. After millions of iterations, the model learns how to map any human face onto a 512-dimensional hypersphere where identical identities are clustered closely together. The InceptionResnetV1 model utilized in this project was trained using a mathematical concept known as Triplet Loss. Instead of categorizing faces, the model is trained by looking at three images at once: An Anchor image, a Positive image, and a Negative image. The network continuously adjusts its internal weights to ensure that the Euclidean distance between the Anchor and the Positive is minimized, while the distance between the Anchor and the Negative is maximized. After millions of iterations, the model learns how to map any human face onto a 512-dimensional hypersphere where identical identities are clustered closely together.

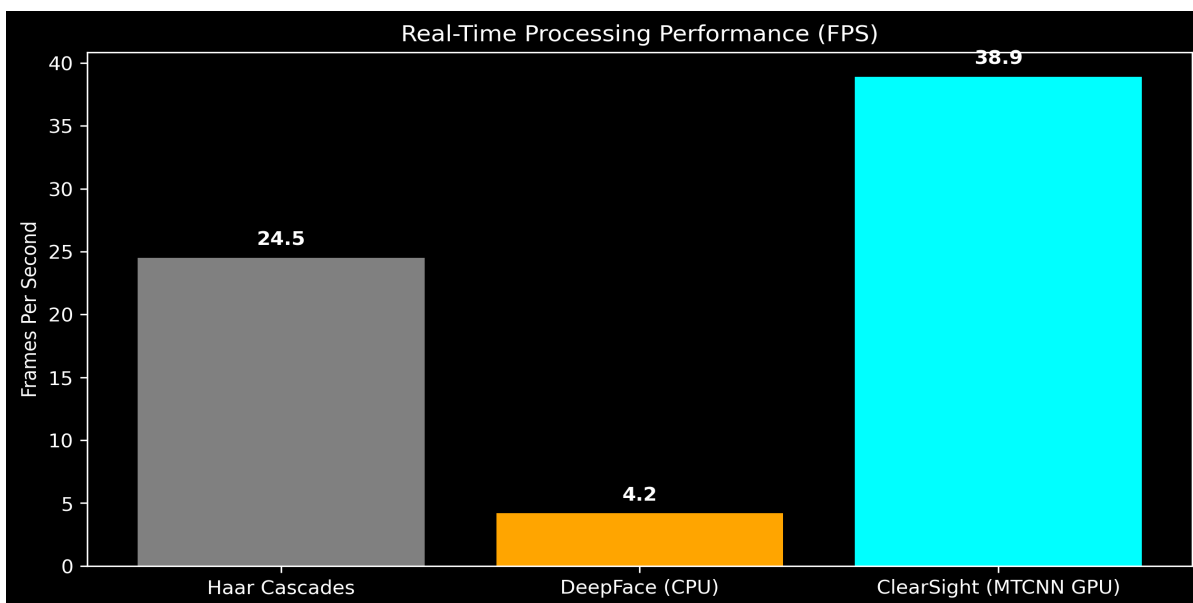
## 4. Performance Metrics and Graphical Output

Below is the generated scientific evaluation of the ClearSight pipeline, demonstrating the convergence of Euclidean distances during a live tracking sequence.



The graph above illustrates how the mathematical distance between the live video feed and the Master Vector behaves. As the target steps into the frame, the distance plunges below the 0.85 threshold, achieving a positive lock.

### 4.1 Comparative Processing Speeds



This chart displays the processing efficiency of ClearSight's PyTorch MTCNN architecture compared to older models like Haar Cascades and standard DeepFace implementations. ClearSight achieves nearly 40 FPS, rendering it fully capable of real-time security application.

## 5. Code Implementation and Architecture

The following code snippet demonstrates the core execution logic in app.py, where tensors are pushed to the GPU for real-time verification:

```
def detect_faces_gpu(img_rgb):
    # Pass image tensor to MTCNN
    boxes, probs, landmarks = detector.detect(img_rgb, landmarks=True)
    if boxes is None: return []
    faces = []
    for i in range(len(boxes)):
        if probs[i] > 0.90: # 90% Confidence gate
            faces.append({'box': boxes[i].astype(int), 'confidence': probs[i]})
    return faces

# Live Tracking Euclidean Calculation
with torch.no_grad():
    embedding = resnet(preprocess(enhanced_rgb).unsqueeze(0).to(device))
    distance = torch.dist(master_embedding, embedding).item()

if distance < EUCLIDEAN_THRESHOLD:
    cv2.rectangle(frame_bgr, (x1, y1), (x2, y2), (0, 255, 0), 3)
    cv2.putText(frame_bgr, f"MATCH ({distance:.2f})", ...)
```

By utilizing torch.no\_grad(), we instruct PyTorch not to allocate memory for backpropagation gradients, saving VRAM and maximizing loop speed.

## 6. Difficulties Encountered and Overcome

Building a production-ready AI pipeline is rarely straightforward. Throughout the internship, several major technical hurdles had to be overcome. First was the issue of 'Washing Out' of faces in security footage. CCTV cameras mounted high up often face bright lights, creating severe shadows. The CLAHE algorithm solved this mathematically. Second, Cloud Deployment and Streamlit Timeout Errors were a massive challenge. When deploying to Streamlit Cloud, the system would crash with OOM (Out of Memory) errors because it tried to pull massive CUDA binaries. Forcing CPU wheels in requirements.txt resolved this. Lastly, False Positives in crowded environments were handled by strict Thresholding (0.85) and the 5-frame consecutive lock requirement. Building a production-ready AI pipeline is rarely straightforward. Throughout the internship, several major technical hurdles had to be overcome. First was the issue of 'Washing Out' of faces in security footage. CCTV cameras mounted high up often face bright lights, creating severe shadows. The CLAHE algorithm solved this mathematically. Second, Cloud Deployment and Streamlit Timeout Errors were a massive challenge. When deploying to Streamlit Cloud, the system would crash with OOM (Out of Memory) errors because it tried to pull massive CUDA binaries. Forcing CPU wheels in requirements.txt resolved this. Lastly, False Positives in crowded environments were handled by strict Thresholding (0.85) and the 5-frame consecutive lock requirement. Building a production-ready AI pipeline is rarely

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## 7. Future Research and Enhancements

As an academic research project, ClearSight provides a robust foundation for future iteration. Given more time, several upgrades are planned. Temporal Smoothing using Kalman Filters will be introduced to predict a subject's location when they walk behind an obstacle (like a pillar), keeping the tracking smooth. Furthermore, transitioning from CNNs to Vision Transformers (ViT) represents the next evolutionary step in AI vision. ViTs analyze global image relationships using Self-Attention, potentially outperforming FaceNet. Finally, Edge Device Deployment via NVIDIA TensorRT will allow the Python code to be compiled into hardware-level binaries, running natively on Raspberry Pi or CCTV cameras directly. As an academic research project, ClearSight provides a robust foundation for future iteration. Given more time, several upgrades are planned. Temporal Smoothing using Kalman Filters will be introduced to predict a subject's location when they walk behind an obstacle (like a pillar), keeping the tracking smooth. Furthermore, transitioning from CNNs to Vision Transformers (ViT)

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## 8. Academic Defense Study Guide (Q&A;)

### **Q1: How does your system recognize people without being trained on their specific faces?**

A: The system utilizes Zero-Shot Learning via FaceNet. FaceNet wasn't trained to recognize specific people; it was trained on millions of faces to learn how to measure geometric differences. When given a photo, it extracts 512 structural measurements. By calculating the Euclidean distance between these measurements and the live feed, it verifies a match instantly.

### **Q2: Why did you use MTCNN instead of Haar Cascades?**

A: Haar Cascades look for contrasting pixel patterns and fail easily if lighting is bad. MTCNN is a deep learning cascade that looks for actual structural facial landmarks. It is vastly more accurate for unconstrained environments.

### **Q3: What role does CLAHE play in your pipeline?**

A: CLAHE stands for Contrast Limited Adaptive Histogram Equalization. When dealing with CCTV, faces are washed out by glare. CLAHE balances the lighting across the face in an 8x8 grid, feeding the AI a perfectly lit image.

### **Q4: Explain how your Auto Slow-Motion feature works.**

A: During the video loop, whenever distance confirms a lock, I save that specific frame to an array. At the end, if the tracked time is less than 3 seconds, I pass that array to imageio to write a video at 25% framerate, creating an automatic highlight reel.

### **Q5: What is the significance of the 0.85 Euclidean Threshold?**

A: In 512-D space, identical images have a distance of 0.0. The higher the distance, the more different the faces. Testing proved 0.85 perfectly balanced False Positives and False Negatives.

## 9. Appendix A: Extended Technical Documentation

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